

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: HMT500(260313A)_connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: HMT500(260313A)_power.kicad_sch

3. MCU

STM32G0B1CCT3, SWD, reset, internal temperature sensor

File: HMT500(260313A)_mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 2-wire (ADS1220)

File: HMT500(260313A)_measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: HMT500(260313A)_analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: HMT500(260313A)_rs485.kicad_sch

DOTECH HMT500(260313) – Oil Moisture Transmitter – Schematic v0.7 – Project No. HMT500(260313A)

File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A)

v0.5: J1 = JST GH 1.25 mm 8P side entry (SM08B-GHS-TB), harness W-2 to M12 8P; J5 chassis spring contact (was M12 shell pin).

v0.4: J3 = JST SH 1.0 mm 4P (SM04B-SRSS-TB side entry, v0.5) for sensor harness W-1 direct from HTX99R (no feedthrough, mech Rev F); Pt1000 2-wire (Kelvin split at J3 pads).

v0.3: MCU STM32G0B1CCT3 (LQFP48, -40..125 C, in-house part); TPS26600PWP official pinout; single axial PCB 57x23 (mech Rev C).

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-30 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.

v0.6: parts cleanup (-16: pull-ups R11-15, LED D3/R16, RS-485 R60/61 D60/61, FB2 C19 C11 C2 C6); PCB temperature = ADS1220 + MCU internal sensors.

v0.7: DAC8760 datasheet pinout (HTSSOP-24 PWP); +VSENSE via OPA197 follower (U12/U13); external ISET-R 15k 0.1%; DACs on SPI2 daisy chain, common DAC_LATCH.

VERIFY: LMR36006, PCAP04, TPS26611 pin numbers are placeholders (datasheets not accessible when drawn). TPS2660: RTN link R22 to confirm.

TBD footprints: J5 chassis contact, CMC, GDT, buck inductor, LMR36006, PCAP04, TPS26611.

표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울임 글자 (+3V3, VIN_P)

신호 네트 = 테두리 있는 라벨 (SPL_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음

글자 겹침: 생성기가 부품 몸체·판·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리

VERIFY=YES parts: pin numbers LMR36006/PCAP04/TPS26611; TPS2660 R22

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter, RS-485 + 2x V/I output

DOTECH Co., Ltd.

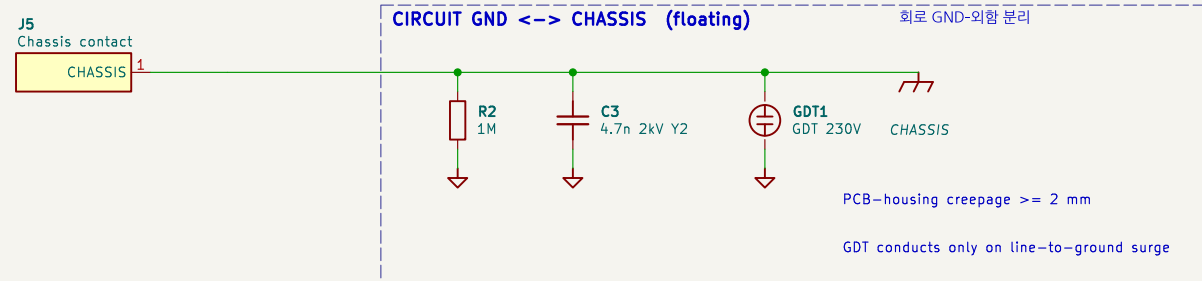
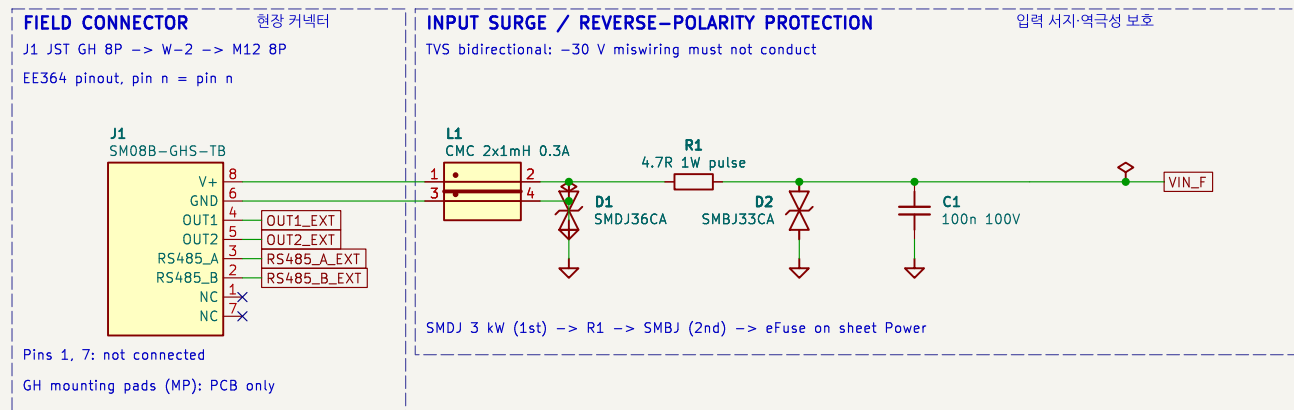
Sheet: /

File: HMT500(260313A).kicad_sch

Title: HMT500(260313A) Top

Size: A3Date: 2026-09-29Rev: 0.7

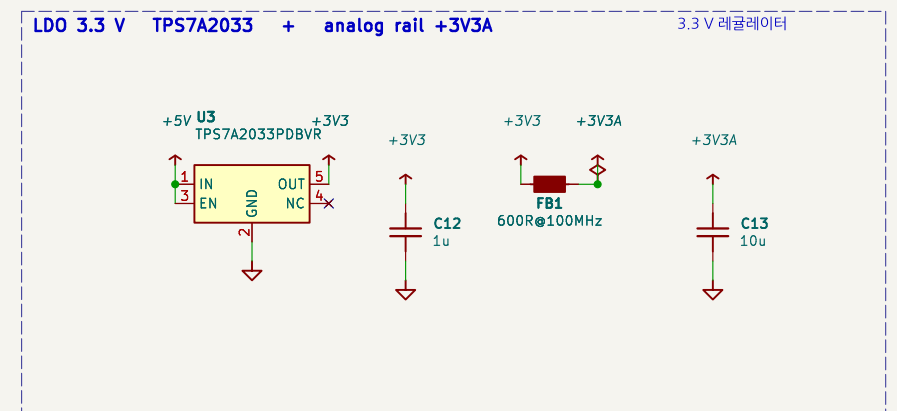
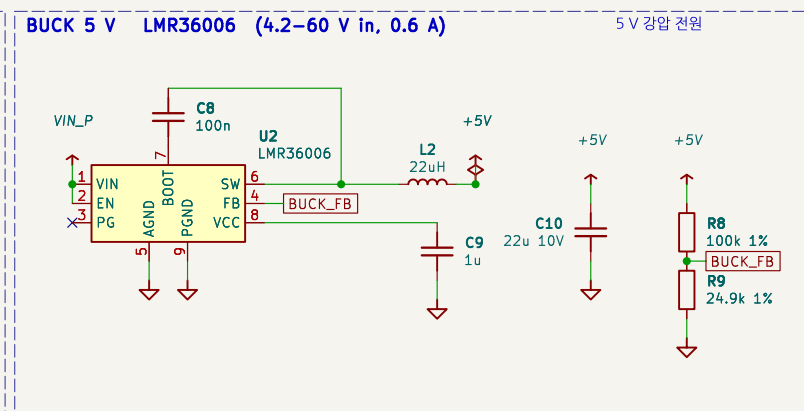
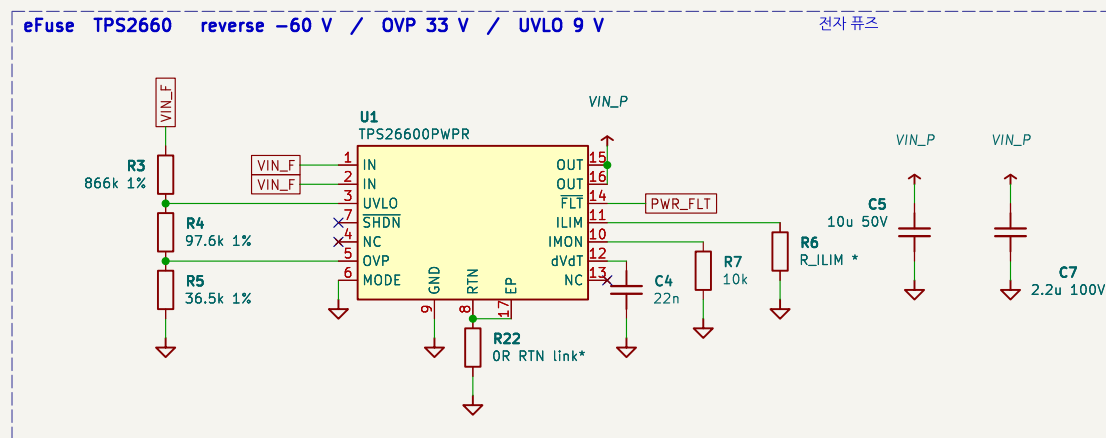
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4Id: 1/7



주요 부품 — 역할·특징·사양			
부품번호	부품	역할	특징·사양
J1	SM08B—GHS—TB	현장 커넥터 연결	JST GH 1.25 mm 8P 옆 삽입, 하네스 W-2 → M12 8P, 핀 n = M12 핀 n
L1	CMC 2x1mH 0.3A	전원선 공통모드 노이즈 차단	2 × 1 mH, 0.3 A — 전도 방출-전도 내성 대책
D1	SMDJ36CA	입력 서지 1단 흡수	TVS 양방향 3000 V, 36 V — -30 V 오결선에 대응 한 함
R1	4.7R 1W pulse	1단·2단 서지 분담	MELF 1 W 펄스 내량 저함
D2	SMBJ33CA	입력 서지 2단 클램프	TVS 양방향 600 V, 33 V — eFuse 정격 이하로 제한
GDT1	GDT 230V	회로 GND-외함 서지 방전	230 V 방전관, 선-대지 서지 때만 도통 (평소 절연)
C3, R2	4.7n Y2 / 1M	GND-외함 고주파 결합	Y2 안전 콘덴서 + 1 MΩ 정전기 방전 경로
J5	Chassis contact	PCB-하우징 겹치 접점	SMD 스프링 접점, PCB 가장자리 → 하우징 Ø27 내면

* 표시 값은 데이터시트로 확인

VERIFY=YES parts: pin numbers LMR36006/PCAP04/TPS26611; TPS2660 R22 Design notes: docs/hw/circuit-design.md Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter, RS-485 + 2x V/I outputs DOTEC Co., Ltd.	
Sheet: /1. Connector & input protection/ File: HMT500(260313A)_connector.kicad_sch	
Title: HMT500(260313A) Connector & input protection	
Size: A3 KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	Date: 2026-09-29 Rev: 0.7 Id: 2/7

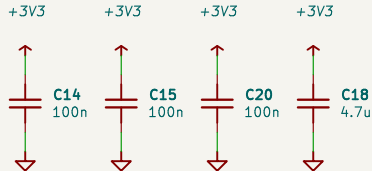


UVLO = $1.2 \text{ V} \times (R3+R4+R5)/(R4+R5) = 8.9 \text{ V}$
 OVP = $1.2 \text{ V} \times (R3+R4+R5)/R5 = 32.9 \text{ V}$ (Vref 1.2 V: VERIFY)
 * R6 sets current limit 150 mA (formula in TPS2660 datasheet)
 *R22 RTN link: verify vs TPS2660 datasheet – RTN tied to GND disables reverse–polarity protection

$$V_{out} = 1.0 \text{ V} \times (1 + R_8/R_9) = 5.0 \text{ V} \quad (V_{ref} 1.0 \text{ V: VERIFY})$$

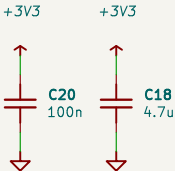
주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
U1	TPS26600PWR	전차 퓨즈 (입력 보호)	역극성 -60 V 차단, 과전압 33 V-저전압 9 V (R3-R5), 전류 제한 (R6)*, 돌입 제한 (C4), 고강 출력 FLT
U2	LMR36006	5 V 강압 전원 (백)	입력 4.2~60 V, 출력 0.6 A 동기식, 무부하 소비전류 작음*, 출력 = 1 V × (1 + R8/R9)*
L2	22μH	백 출력 인덕터	4 × 4 mm 차폐형, 포화 전류 0.8 A 이상으로 선정
U3	TPS7A2033PDBVR	3.3 V 저잡음 레귤레이터	300 mA, 저잡음-높은 PSRR* → 백 리플 제거, SOT-23-5
FB1	600R@100MHz	아날로그 전원 +3V3A 분리	페라이트 비드 + C13 10 uF → PCAP04-AD51220 전원
* 표시 값은 데이터시트로 확인			

MCU DECOUPLING



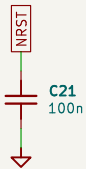
100 nF at VDD, VBAT, VREF+ (C20); 4.7 uF bulk

MCU 바이패스



RESET

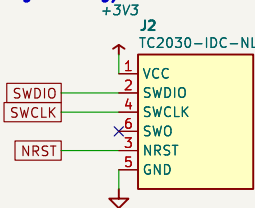
리셋



Fault lines: MCU pull-ups
(PB3-4, PB11, PC6-7 internal)

SWD (production programming)

생산 프로그래밍



MCU NOTES (STM32G0B1)

MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1

-> boot from flash (RS-485 bootloader 0x08000000).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U4	STM32G0B1CCT3	제어-통신-보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-125 °C, LQFP48, 내부 온도센서 (PCB 온도), SPI1 = 측정 · SPI2 = DAC 전용
J2	TC2030-IDC-NL	생산 때 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유

* 표시 값은 데이터시트로 확인

VERIFY=YES parts: pin numbers LMR36006/PCAP04/TPS26611; TPS2660 R22

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter, RS-485 + 2x V/I output

DOTECH Co., Ltd.

Sheet: /3. MCU/

File: HMT500(260313A)_mcu.kicad_sch

Title: HMT500(260313A) MCU

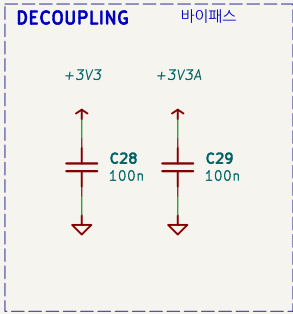
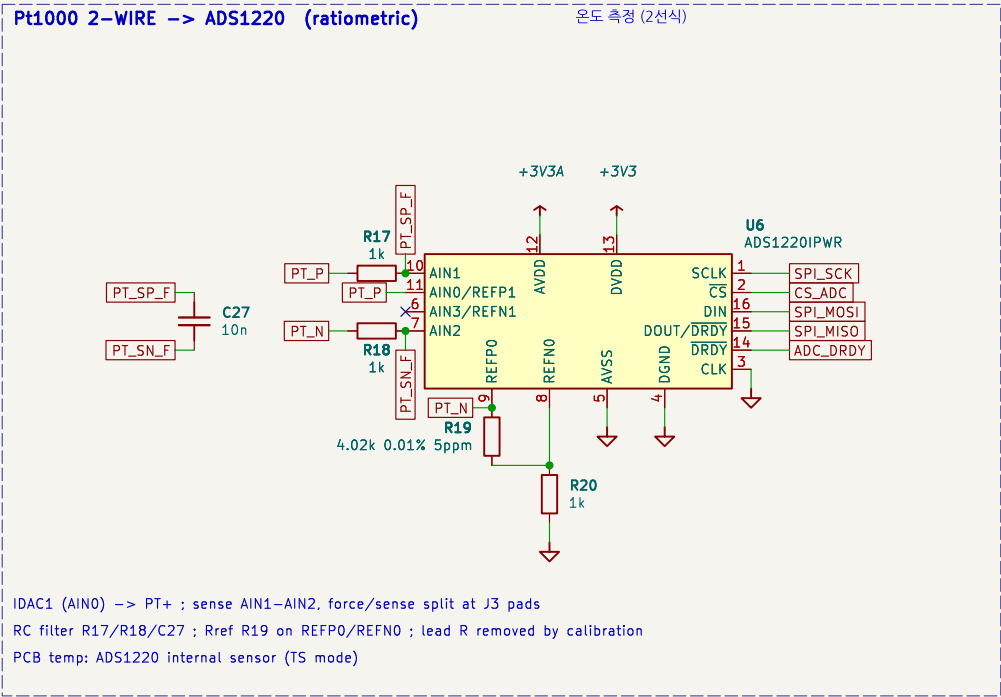
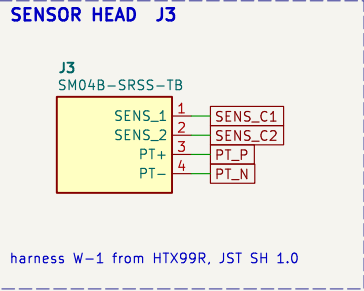
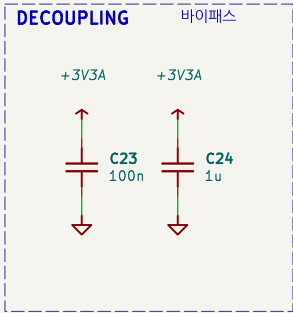
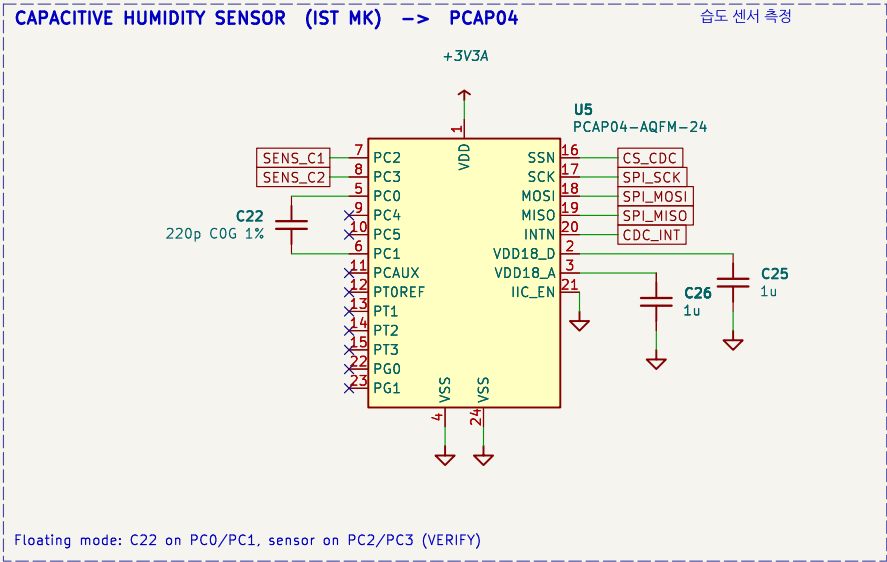
Size: A3

Date: 2026-09-29

Rev: 0.7

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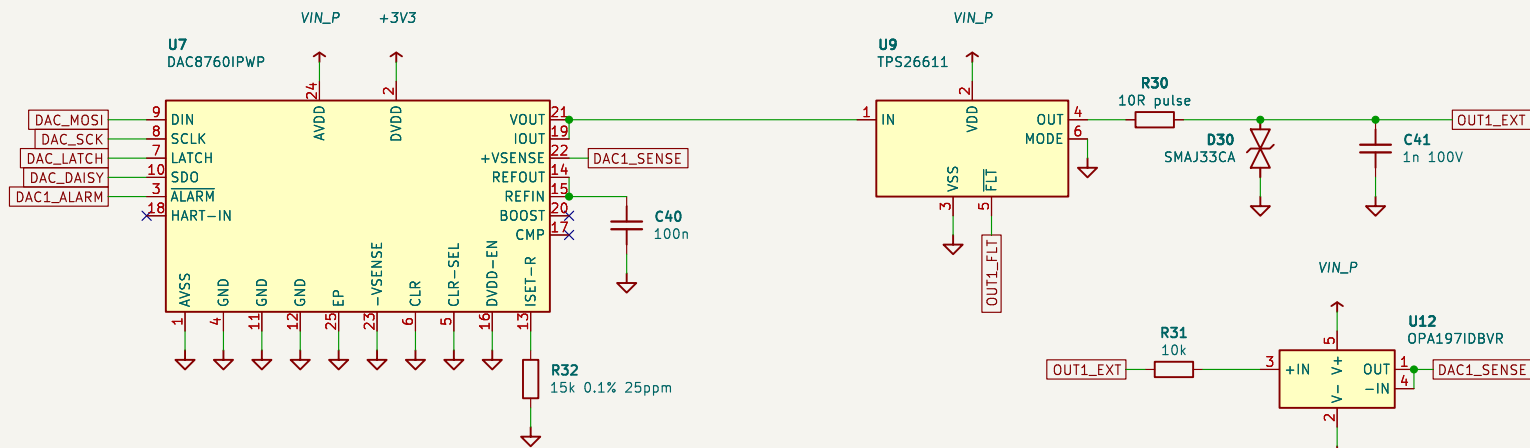
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주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
J3	SM04B-SR55-TB	센서 하네스 W-1 연결	JST SH 1.0 mm 4P 옆 삽입, 1-2 = MK33, 3-4 = Pt1000
U5	PCAP04-AQFM-24	습도 센서 정전용량 측정	정전용량-디지털 변환, 기준 C와 비율 측정, 내부 DSP, SPI*
C22	220p COG 1%	PCAP04 기준 용량	MK33-W mini (200 pF)와 비슷한 값, COG (온도 변화 거의 없음)
U6	ADS1220IPWR	Pt1000 온도 측정	24비트 ADC, PGA, 여가 전류 IDAC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도)
R19	4.02k 0.01% 5ppm	Pt1000 비율 측정 기준저항	IDAC 전류 오차 상쇄, 온도계수 5 ppm/°C
R17, R18, C27	1k / 10n	입력 RC 필터	차동-공통모드 노이즈 제거

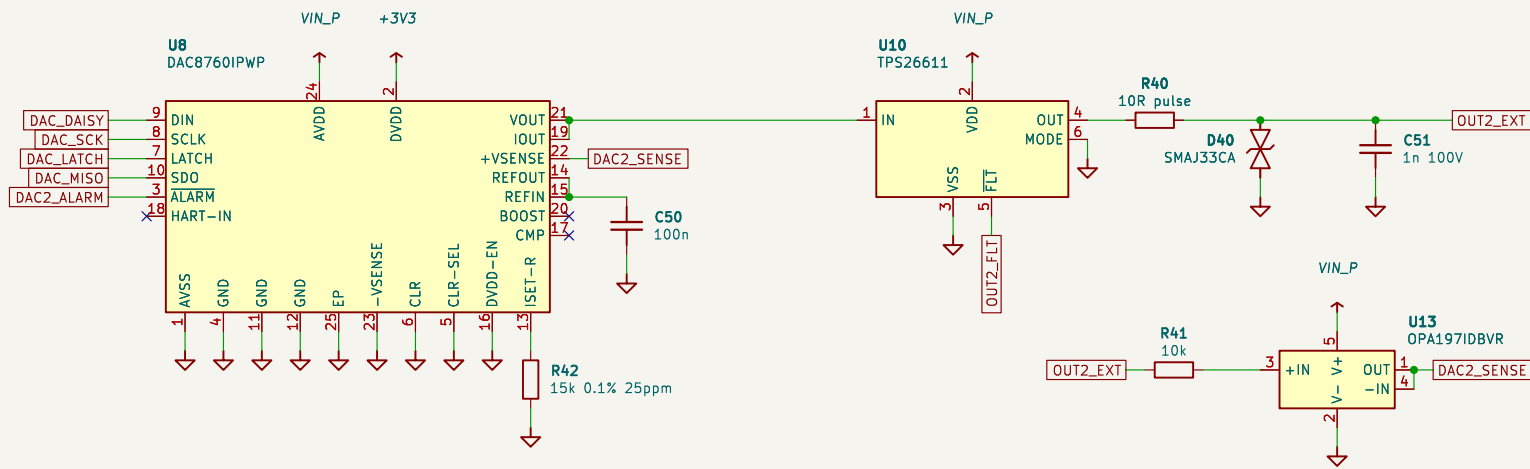
* 표시 값은 데이터시트로 확인

ANALOG OUTPUT CH1 - V/I selectable, miswiring protected -> connector pin 4



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U12 OPA197 follower of terminal OUT1_EXT (after R30) via R31:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R32 external 15k 0.1%.

ANALOG OUTPUT CH2 - V/I selectable, miswiring protected -> connector pin 5



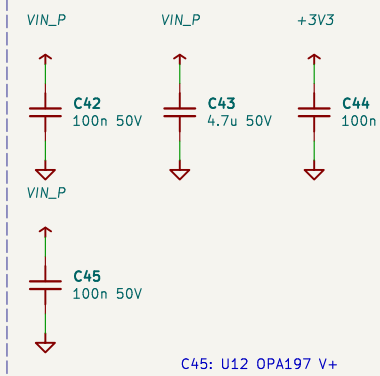
VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U13 OPA197 follower of terminal OUT2_EXT (after R40) via R41:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R42 external 15k 0.1%.

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U7, U8	DAC8760IPWP	아날로그 출력 1·2	16비트, 4-20 / 0-20 / 0-24 mA · 0-5 / 0-10 V 선택, 내부 기준, 고정 알람 — 대체 DAC7760/8750/7750
U9, U10	TPS26611	출력 오걸선 보호	출력 단자 과전압·역전압 차단, 전류 제한, 고정 출력 FLT*
R30, R40	10R pulse (2512)	서지 전류 제한	출력 보호기와 TVS 사이 직렬
D30, D40	SMAJ33CA	출력 서지 클램프	TVS 양방향 400 V, 33 V
C41, C51	1n 100V	출력 고주파 필터	케이ابل로 들어오는 RF 억제
U12, U13	OPA1971DBVR	+VSENSE 버퍼 (이득 1)	36 V, 임출력 레일투레일, 오프셋 100 μ V — 단자 전압을 +VSENSE로, 전류 모드 누설 0
R31, R41	10k	버퍼 입력 보호	R30-R40 뒤 단자 전압을 OPA197 + 입력으로 (서지 전류 제한)
R32, R42	15k 0.1% 25ppm	외부 ISET-R (전류 기준)	내부 저항보다 온도 드리프트 작음 → 전류 출력 정확도
R33	100k	데이터 인 폴다운	U7 SDO 켜기 전 U8에 NOP (0x00) 입력

* 표시 값은 데이터시트로 확인

CH1 DECOUPLING



DAC SELECTION (decision #10)

U7/U8 fitted: DAC8760 (16-bit, I + V output).

Drop-in alternates (same pinout, no layout change):

DAC7760 12-bit, I + V (cost-down, V/I kept)

DAC8750 16-bit, I only 4-20/0-20/0-24 mA (EE364-equiv.)

DAC7750 12-bit, I only (-5 μ A/step vs EE364 2 μ A)

Current-only (x750): VOUT/+VSENSE/-VSENSE absent –

check pin handling before fitting (VERIFY).

DAC SPI – SPI2 daisy chain

SPI2 for the DACs only (PB13 SCK, PB14 MISO, PB15 MOSI).

DAC_MOSI -> U7 DIN, U7 SDO -> DAC_DAI5Y -> U8 DIN,

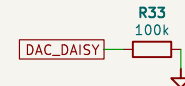
U8 SDO -> DAC_MISO (readback / fault status).

One 48-bit frame (U8 word first), then one DAC_LATCH

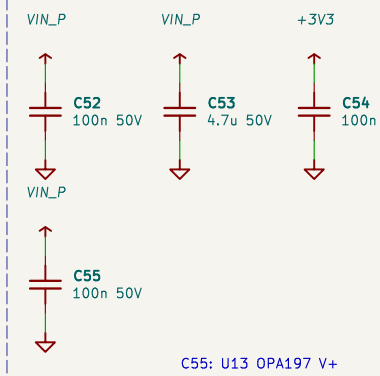
(PB10) edge: both channels update together.

Init: set DCEN in U7 (24-bit), then in U8 (48-bit).

R33: U8 shifts in NOP (0x00) until U7 SDO is enabled.



CH2 DECOUPLING



VERIFY=YES parts: pin numbers LMR36006/PCAP04/TPS26611; TPS2660 R22

Design notes: docs/hw/circuit-design.md

Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter, RS-485 + 2x V/I output

DO TECH Co., Ltd.

Sheet: /5. Analog outputs/

File: HMT500(260313A)_analog_out.kicad_sch

Title: HMT500(260313A) Analog outputs

Size: A3

Date: 2026-09-29

Rev: 0.7

Size: AS	Date: 2020-09-29
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